



Financial Institutions & Markets- ENT 309

PPT #2, IITD, Summer, 2026

Agenda

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- Types of Debt Instruments
- Comparing two loan offers
- Discount Rate, NPV, IRR.
- Bonds
- Yield To Maturity (YTM)
- Discount Rate, YTM, Interest Rate
- Change in Bond Prices with Interest Rate
- Perpetuity Bond & Discount Bond
- Real & Nominal Interest Rates

Debt Market Products



Debt Market

Instrument	Issuer	Maturity	Tradable	Secured?	Typical Use	Market
CD	Banks	Short	Yes	Bank liability	Funding	Money Market
Repo	Banks/RBI	Very short	Yes	Yes	Liquidity	
T-Bills	Govt	<1 year	Yes	Sovereign	Liquidity	
CP	Corporates	<1 year	Yes	No	Working capital	



Debt Market

Instrument	Issuer	Maturity	Tradable	Secured?	Typical Use	Market
NCD	Corporates	Medium/Long	Yes	Sometimes	Expansion	Capital Market
ECB	Foreign lenders	Medium/Long	Sometimes	Varies	Foreign funding	
G-Secs	Govt	Long	Yes	Sovereign	Govt borrowing	
SDLs	State Govt	Long	Yes	Quasi-sovereign	State borrowing	



Debt Market

Instrument	Issuer	Maturity	Tradable	Secured?	Typical Use	Market
Bank loan	Banks	Flexible	No	Often	Projects	Bank Debt
FD	Banks	Flexible	No	Deposit claim	Savings	

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Time Value of Money



Which Deal is Better?

- You have Rs. 15 lacs with you. Ram & Shyam want to borrow it from you on interest. Who will you give your money to?
- How will you make the decision?

	Ram	Shyam
LOAN GIVEN (INR)	1500000	1500000
REPAID OVER (YEARS)	5	5
TOTAL AMOUNT REPAID (INR)	22,50,000	23,50,000

Which Deal is Better?



REPAYMENTS/CASH FLOW CF	Ram	Shyam
Year 1	450000	100000
Year 2	450000	150000
Year 3	450000	450000
Year 4	450000	650000
Year 5	450000	1000000
TOTAL REPAID	22,50,000	23,50,000

Compare Net Present Value of the Two Cash Flows

Which Deal is Better?



REPAYMENTS/CASH FLOW CF	Ram	Shyam
Year 1	450000	100000
Year 2	450000	150000
Year 3	450000	450000
Year 4	450000	650000
Year 5	450000	1000000
TOTAL REPAID	22,50,000	23,50,000
Net Present Value NPV	17,50,343	16,75,885
Discounting Rate (i)	9%	9%

Present Value

$$PV = \frac{CF}{(1 + i)^n}$$

Discount Rate



- What determines the Discount Rate? Discounting Factor.
- It reflects the **Time-Value of money**.
- Risk Free Rate =?
- Sovereign/ Gov 10 Year Bond Yield. ~ 7% currently.

Which Deal is Better?



- Find NPV of Case 1 and Case 2. Which is better?
- What is IRR?
- Find IRR of Case 1.
- Use IRR as Discount Rate. What do you observe?
- At Discount Rate = IRR, what is the NPV?
- **REMEMBER IRR USED IN THE CONTEXT OF LOANS ONLY AND NOT BONDS.**
- **CONCEPTUALLY IRR IS SAME AS YIELD TO MATURITY FOR BONDS**

Discount Rate, NPV, IRR



Discount Rate	9.000%
Loan Amount	1500000

=NPV(rate, CF1, CF2.....)

Year	Ram	Shyam	Discounting Factor	Disc CF Ram	Disc CF Shyam
1	450000	100000	0.91743119	412844	91743
2	450000	150000	0.84167999	378756	126252
3	450000	450000	0.77218348	347483	347483
4	450000	650000	0.70842521	318791	460476
5	450000	1000000	0.64993139	292469	649931
Total Repaid	2250000	2350000			
NPV of Installments	₹ 17,50,343	₹ 16,75,885		1750343	1675885

Discount Rate, NPV, IRR



Time Period	CF Case 1
0	-1500000
1	450000
2	450000
3	450000
4	450000
5	450000
IRR	15.238%

Discount Rate	15.238%
Loan Amount	1500000
Year	CASE 1
1	450000
2	450000
3	450000
4	450000
5	450000
Total Repaid	2250000
NPV of Installments	₹ 15,00,008

Use IRR as
Discount Rate.

What
happens?

Repayment Schedule, IRR



		REPAYMENT SCHEDULE		
Time Period	CF Case 1	Principal	Interest @15.238%	Principal Remaining
0	-1500000			1500000
1	450000	221426	228574	1278574
2	450000	255168	194832	1023406
3	450000	294051	155949	729355
4	450000	338859	111141	390495
5	450000	390495	59505	0
IRR	15.238%	1500000	750000	

Reducing Balance Interest Calculation for a Loan of Rs 15 lacs Repaid in Yearly Instalments of Rs 4.5 lacs Over 5 Years.

About Bonds

Fixed-Income Security or Bond



- Debt instruments. Pay a fixed amount of interest—in the form of coupon payments—to investors.
- Interest payments made annually / semiannually / quarterly. Principal invested returns to the investor at maturity.
- Bonds are the most common form of fixed-income securities.
- Companies raise capital by issuing fixed-income products to investors.



Bond Characteristics

- **Bond:** Security that obligates the ISSUER to make payments to holder over time
- **Face Value or Par Value:** Payment to bondholder at maturity of bond
- **Coupon Rate:** Bond's annual interest payment (%)
- **Maturity Date:** Date on which the bond expires
- **Zero-Coupon Bond:** Pays no coupons, sells at discount, provides only payment of par value at maturity



Cash Flow: 4 Types

- **One time/Bullet Repayment:** (Principal+ Interest) Repaid at Maturity. Compound interest calculation.
- **Fixed Instalment:** Fixed amount. Fixed Frequency. (Part of Principal + Interest). Like a loan EMI.
- **Coupon Bond:** Interest @ coupon rate paid regularly. Face/Par value of bond paid at Maturity.
- **Zero-Coupon Bond/Discount Bond:** Bond sold at a discount. No regular coupon payment. Face/Par value paid at Maturity.

Cash Flow: 4 Types



1	A Gold Loan of INR 2.5 lacs to be repaid after 11 months	A	Fixed Instalment
2	7.26% GOI 2032 Bond	B	Zero Coupon Bond
3	Mortgage loan of INR 3 crores. EMI of INR 3.0 lacs. For 15 years	C	Bullet Repayment
4	Bond issue price INR 90,000. Face Value INR 1 lac. Maturity after 1 year.	D	Coupon Bond

1- C

2-D

3-A

4-B



Cash Flow Type 1 & 2

- You lend Rs 100 today. After two years, you get back Rs 121 in a bullet payment. What is the interest rate charged?
- $100 = 121/(1+i)^2$. Therefore $i=10\%$
- You have taken a home loan of INR 2 crores. At 7% to be repaid in 20 years. What is the EMI to be paid?
- PMT / Payment function in excel.
- $EMI = ₹ 1,55,060$. $=PMT(Interest\ Rate/12, Years*12, -Principal, 0, 0)$
- If EMI is known,

$Interest\ Rate = RATE(Years*12, EMI, -Principal, 0, 0, Guess)*12$

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Yield to Maturity For a Bond

Yield to Maturity (YTM)



- The interest rate for Bonds is calculated using Yield to Maturity.
- Today's Value/Price of the Bond = Present Value of all the future cash flows.
- Discount Rate used to calculate Present Value = YTM
- **WE WILL USE INTEREST RATE, DISCOUNT RATE AND YTM INTERCHANGABLY AS THE SITUATION DEMANDS.**



YTM for a Bond

- Face or Par Value: Rs 1000
- Coupon: 10%
- Years to Maturity: 10 years.
- Price of the Bond Today=1000
- YTM= r = Discount Rate=?

$$P_0 = \sum_{t=1}^T \frac{C}{(1+r)^t} + \frac{F}{(1+r)^T}$$

Diagram illustrating the Bond Value formula with labels:

- BOND VALUE points to P_0
- Interest Payment (\$) points to C
- Par Value points to F
- Interest Rate points to r
- Number of Periods points to T



YTM for a Bond

- Face or Par Value: Rs 1000
- Coupon: 10%
- Years to Maturity: 10 years.
- Price of the Bond Today= 1200/1100/1000/900/800
- YTM=?
- YTM WILL BE THE IRR AT WHICH PV = DISCOUNTED CASH FLOWS
- **REMEMBER IRR IS NEVER USED IN THE CONTEXT OF BONDS. ONLY YTM OR INTEREST RATE.**

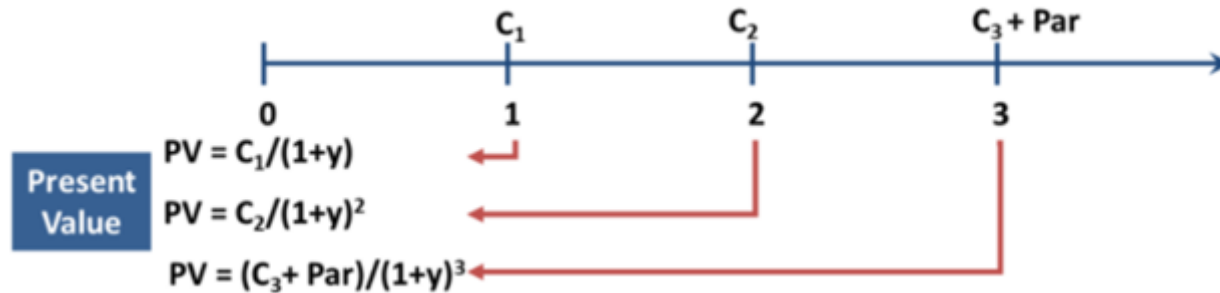


YTM for a Bond

- Price =1200, YTM= 7.13%
- Price=1100, YTM= 8.78%
- Price =1000, YTM=10%
- Price = 900. YTM=11.75%
- Price=800. YTM=13.81%
- What is the relationship between Price, Face Value & YTM as compared to Coupon Rate?
- When current price > Par value: $YTM < \text{Coupon Rate}$
- When current price = Par value: $YTM = \text{Coupon Rate}$
- When current price < Par value: $YTM > \text{Coupon Rate}$



YTM for a Bond



- C = Coupon Rate, Y = YTM, Par = Face Value

ANOTHER WAY OF UNDERSTANDING

- If $C > Y$, Bond trades at a Premium
- If $C < Y$, Bond Trades at a Discount
- If $C = Y$, Bond trades at Par Value/ Face Value



- Discount Rate/YTM/Int Rate is often taken as the risk free rate – The 10 Year Gov Bond Yield
- What happens to bond prices when interest rates change?



3Y

5Y

10Y

MAX



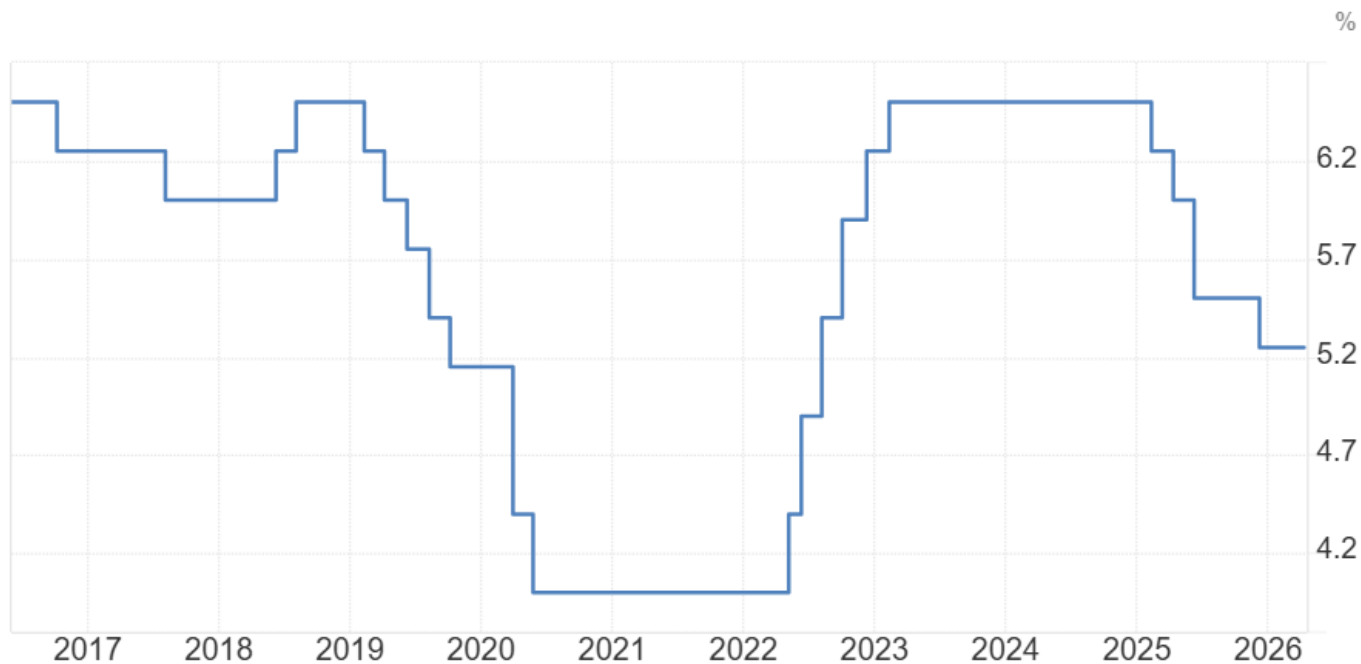
Compare +



Export



API



Reserve Bank of India

Value Chg

Repo Rate Trend in India

- Risk Free Rate for short maturity debt instruments



Change in Bond Price

- What happens to the Price of Bond as YTM rises?
- Price=Present Value PV falls.
- As YTM decreases?
- Price=Present Value PV rises.
- Inversely proportional
- Interest rate increases= YTM increases = Discounting rate increases. All denominators increase.
- And Vice Versa.
- HOW TO EXPLAIN USING ONLY ENGLISH?



Perpetuity Bond

- Bond Price = P
- Coupon = Rs. C per year
- **C paid forever. No expiry date. No payment of par value at maturity.**
- Assume YTM is ' i '
- Derive the relationship between P , C and i .

$$P_c = \frac{C}{i_c}$$

Where

P_c = price of the perpetuity (consol)

C = yearly payment

i_c = yield to maturity of the perpetuity (consol)



Perpetuity Bond

- Perpetuity Bond. Known as AT1 Bonds. Issued by banks.
- Higher risk than normal bonds. Higher coupon than normal bonds.
- No guaranteed payment. And no stock ownership.

Business Standard

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Legal battle over Yes Bank AT1 bond write-off reaches Supreme Court

The dispute stems from the 2020 reconstruction of Yes Bank, when Rs 8,400 crore of AT1 bonds were written down as part of the RBI-led rescue plan

Zero Coupon/ Discount Bond

- Face or Par Value: Rs 1000
- **Coupon: 0%** Maturity: 10 years.
- YTM=8% Price of the Bond= ?
Price= Rs 463.19
- Price of the Bond Today= 600 YTM= ?
YTM= 5.24%
- **Current Price Less Than Par Value. Hence Discount Bond.**
- Relationship between Change in YTM & Change in Price?



Thank You!